Task:

- Classification of the input signal

Categories:

- Noise: 0
- Brushing: 1

Characteristics/Features of the signal:

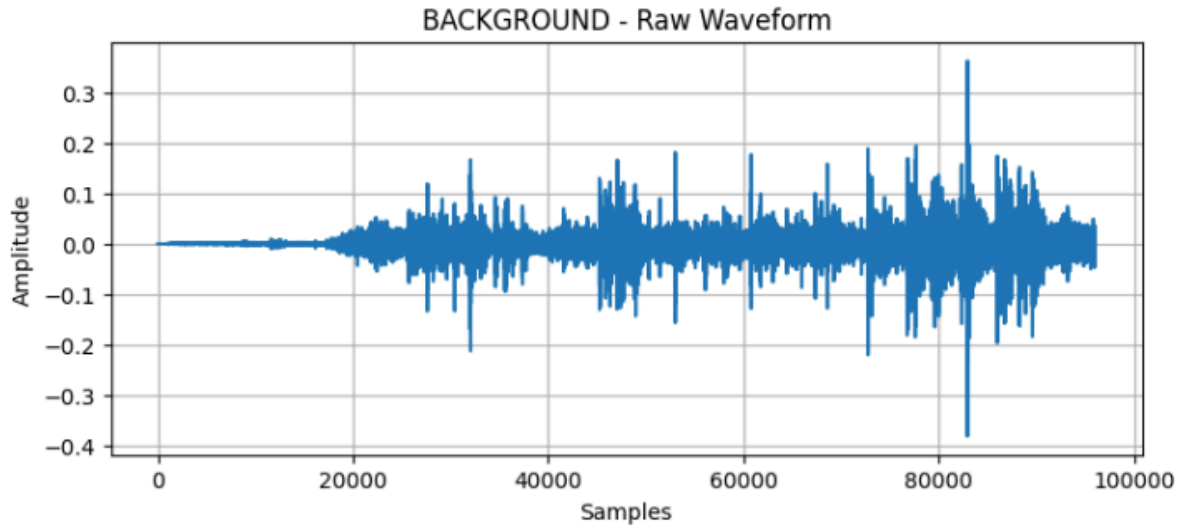
- Amplitude
- Time
- Amplitude/Time

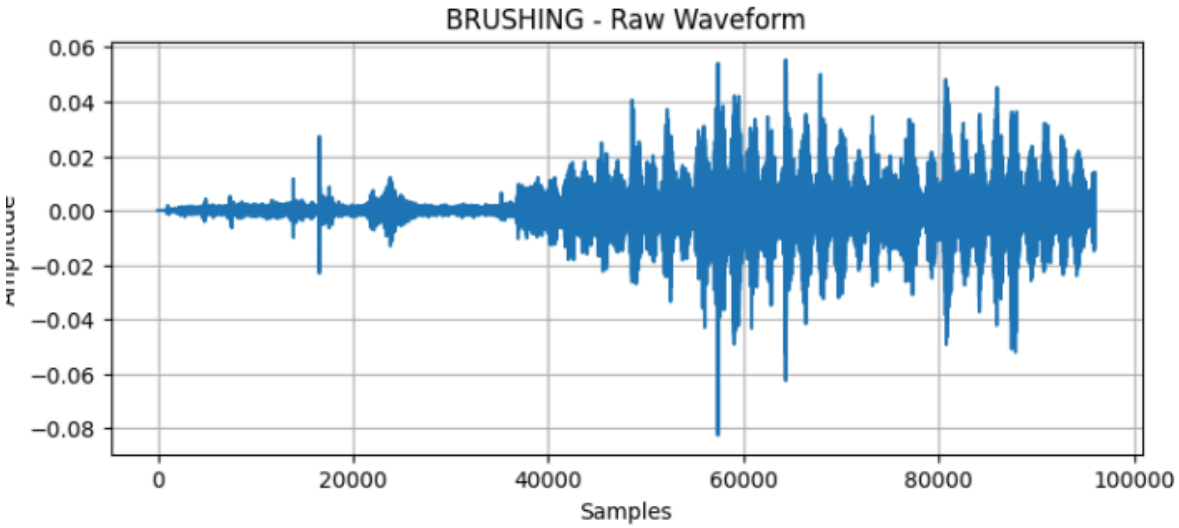
Best algorithms for the best accuracy:

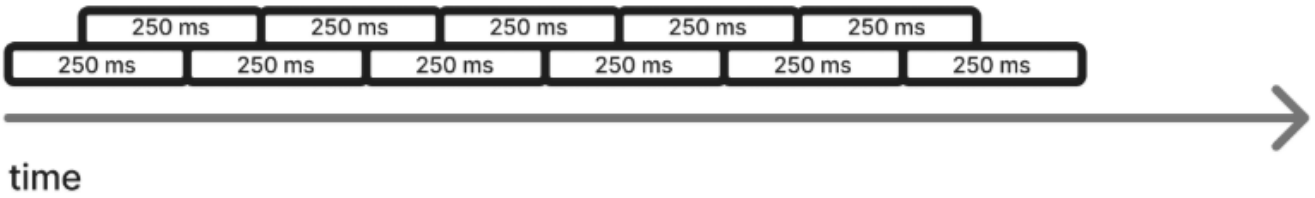
- Classical ML algorithms: Decision Trees or Random Forest
- Deep Networks: simple MLP or CNN

Model requirements:

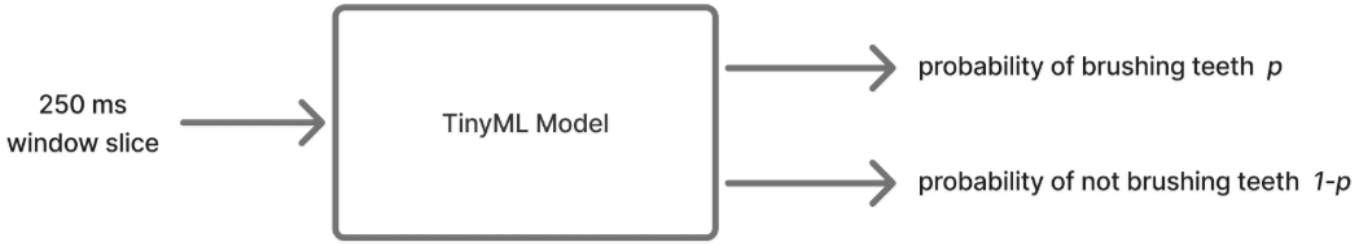
- Storage: up to 1 MB
- RAM: up to 120 kB





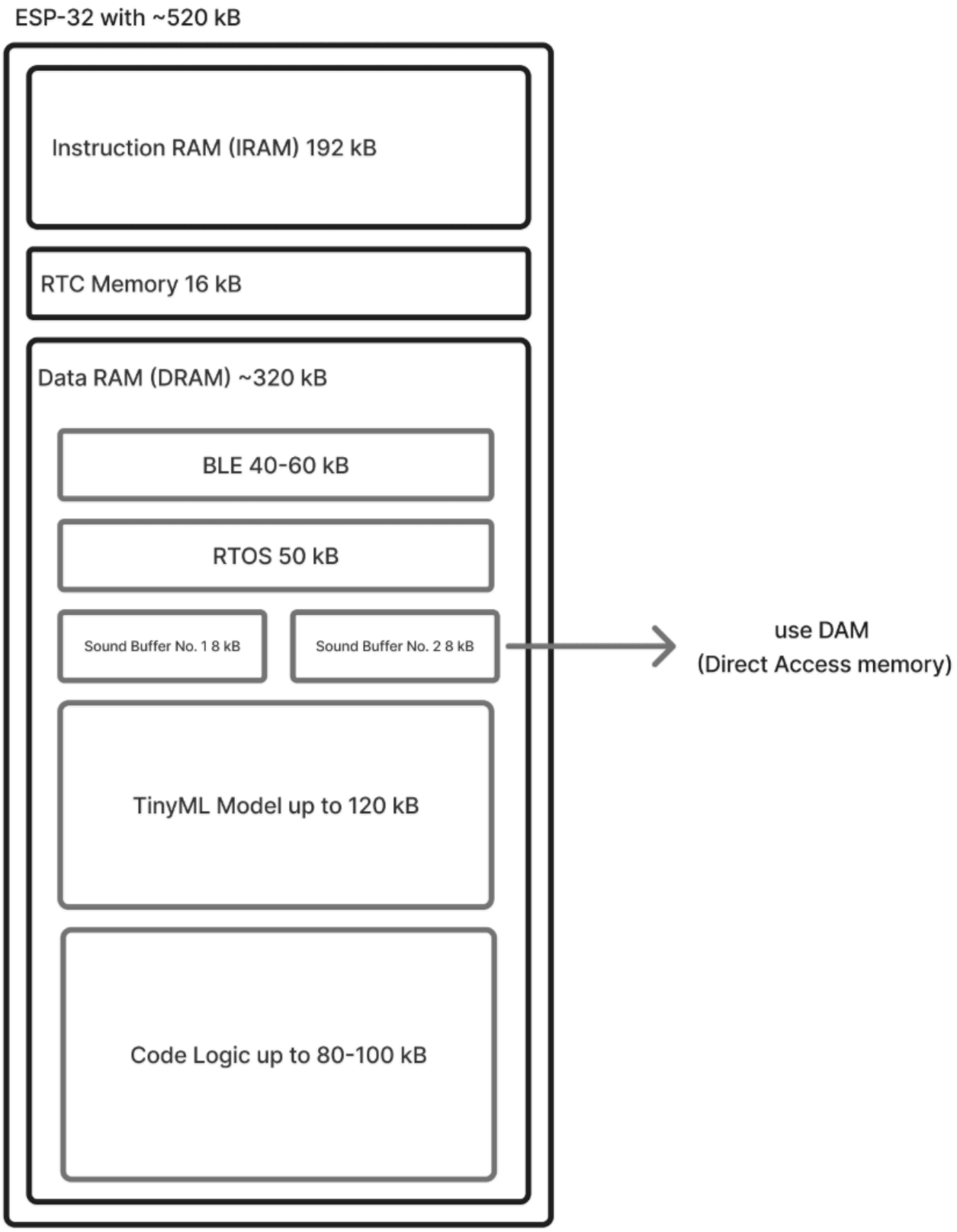


- 4 consecutive positive window results in +1 second of brushing teeth.
- At 16 kHz, the recording will weigh 8 kB and will be saved in sound buffers.
- While processing the first buffer, the next window slice is saved to the second audio buffer, repeating this cycle for the entire teeth-brushing session.
- After saving, we will:
    - *Add here more*
- After processing raw audio, we feed it into the TinyML model and check the output.



Training dataset and process:

- *Add here more about how the dataset will be created and what it will contain*


ESP-32 with ~520 kB
Instruction RAM (IRAM) 192 kB
RTC Memory 16 kB
Data RAM (DRAM) ~320 kB
BLE 40-60 kB
RTOS 50 kB
Sound Buffer No. 1 8 kB
Sound Buffer No. 2 8 kB
use DAM
(Direct Access memory)
TinyML Model up to 120 kB
Code Logic up to 80-100 kB